

# Quark — systematic trading research, done honestly

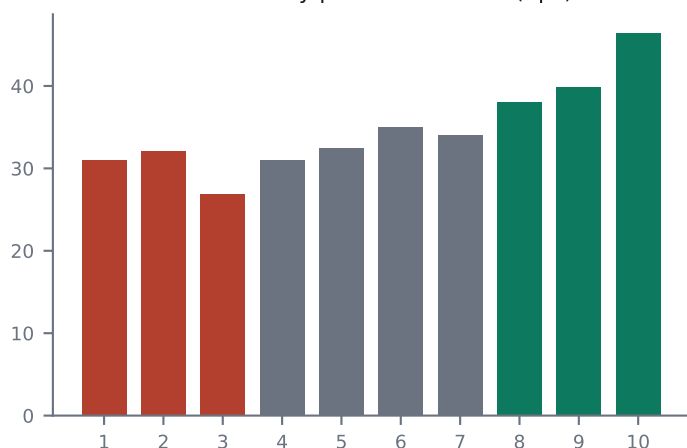
Savitur Swarup · [github.com/Savi-Swar/StrategyBuilder](https://github.com/Savi-Swar/StrategyBuilder)

all results out-of-sample 2012+, net of costs, purged walk-forward, deflated for trial count

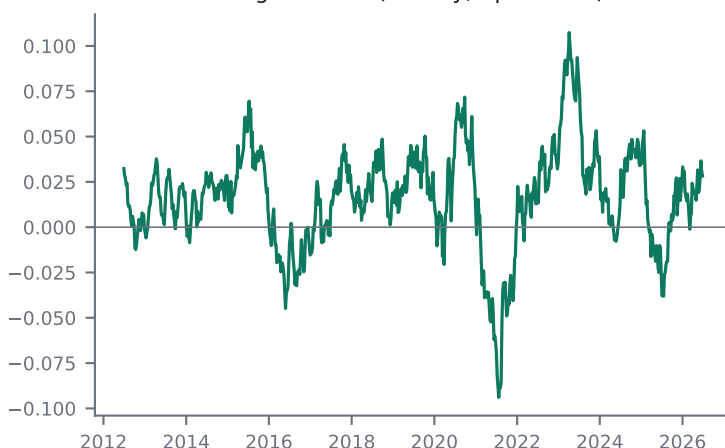
## Headline finding

A gradient-boosted cross-sectional model over ~500 S&P 500 stocks predicts next-week relative returns: weekly rank IC = +0.0172 ( $t = 3.35$ ,  $n = 756$  weekly obs; lag-1 autocorr  $\approx 0$ , Newey-West  $t$  identical), near-monotonic decile spread. Economics are thin after costs — the honest headline is predictive power, not easy money: net Sharpe 0.04 weekly, 0.26 monthly; no-trade bands lift it to 0.13–0.18, monotone across the three registered exit gaps, at ~half the cost drag. Controls: shuffled-label AUC 0.500 / IC  $\approx 0$ ; the long side carries the signal (shorting did not pay).

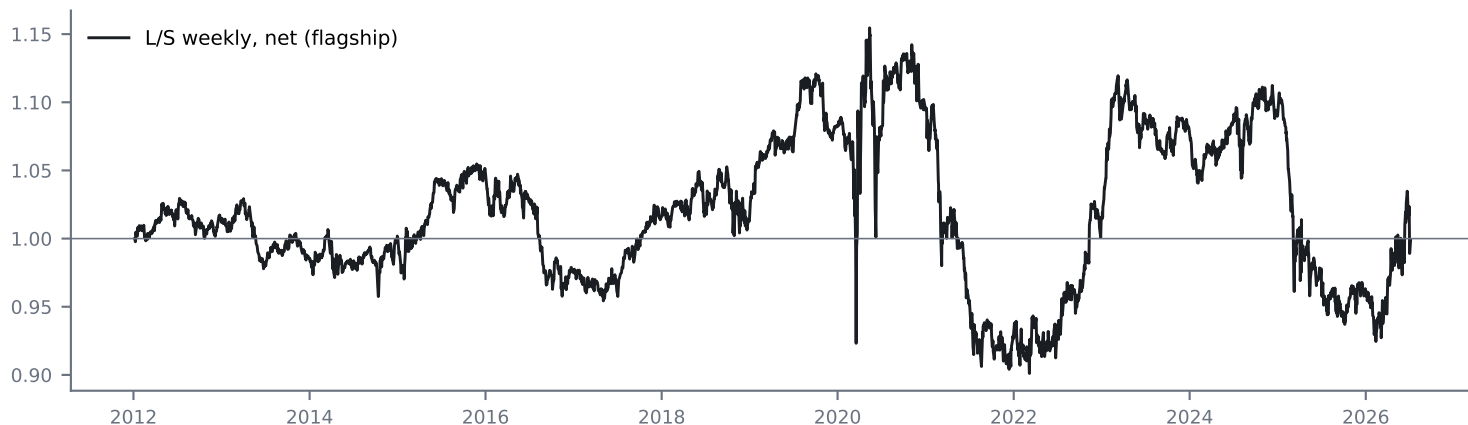
Gross fwd 5d return by prediction decile (bps)



26-week rolling mean IC (weekly, Spearman)



Dollar-neutral extreme-decile long/short — net equity curve



## What failed, reported anyway

Classic indicators (8 counted variants): best = `ma_cross_50_200`, Deflated Sharpe Ratio 0.19 — the best of 8 coin flips. Daily ML timing on 78 multi-asset instruments: AUC 0.518 does not clear costs (net Sharpe -0.44, DSR  $\approx 0$ ). Survivorship bias measured, not just disclaimed: a point-in-time rerun keeps the signal (IC +0.0127,  $t = 2.38$ ) but cuts the decile spread 40% — shipped numbers are upper bounds.

## Why the numbers can be trusted

### Lookahead

positions = signal.shift(lag), lag  $\geq 1$  enforced; unit test: prescient signal earns exactly zero

### Leakage

purged + embargoed walk-forward (purge  $\geq$  horizon); shuffled-label control run for every ML study

### Costs

per-asset-class bps on turnover, entry bar charged; a missing cost rate raises, never defaults to free

### Selection

every variant in a counted registry; best-of-family Sharpe deflated (Bailey & López de Prado DSR)

### Verification

68 unit tests on synthetic fixtures; independent cross-source price verification in the daily job

*Ships as “Vig”: a serverless self-grading terminal — every prediction logged to a ledger and scored after its horizon completes; a live trust gate turns the desk off when the trailing IC decays.*